

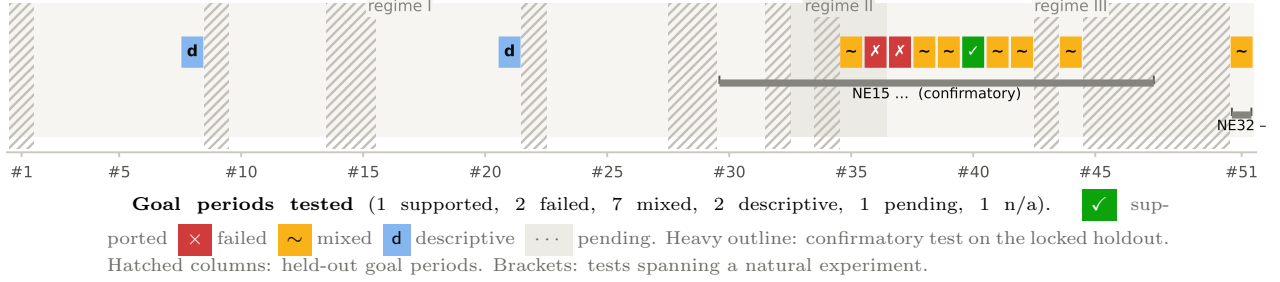
H01: Emergent superagents exist

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 25% Faithfulness 2/5 Usefulness 2/5

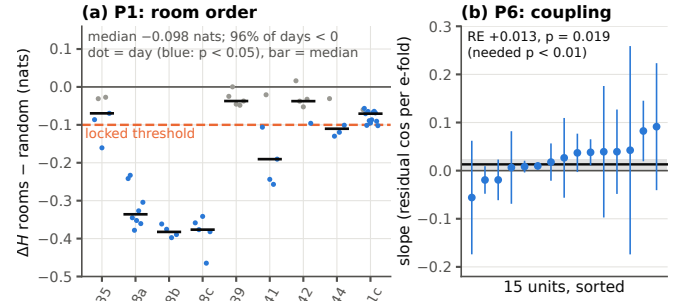
Bottom line. Rooms are measurably more like-minded than random groups, but both locked thresholds were narrowly missed and the order looks more like operator-imposed room fields than an emergent superagent.

level: hypothesis · pipeline checks 85% · updated 2026-10-04



WHAT HAPPENED

Starting set D3.1.a (rooms as ordered “ideological” units) and D3.2 (coupling vs field), tested on 184,713 embedded chat messages and intentions in 16 non-holdout regime II/III units. Rooms had lower meaning-cluster entropy than random same-size groups on 96% of 54 days, but the median gap (-0.098 nats) *narrowly missed* the locked -0.1 . The biggest gaps came where rooms got different instructions (#38: -0.34 to -0.38), i.e. a room field. The exposure slope of residual alignment ($+0.013$ per e-fold) was positive in 12/15 units and beat the rotation null ($p = 0.005$), but its random-effects $p = 0.019$ missed the required 0.01 and it vanished at $n = 16$. Merging rooms (05-04) raised new pairs’ alignment by $+0.18$ ($p = 0.004$, confounded with a goal change). Failed: field dominance (P5 at null level), day-to-day stability (P3), lab order as style (P2), and weak coupling (P9: median $\beta J_0/n = 0.74$). The NE15 holdout test is written but **not run**.



Observables. (a) Each dot is one day’s room-minus-random entropy gap; bars are unit medians. Rooms sit below random on 96% of days, but the pooled median lands just above the locked -0.1 line. The deep gaps (#38a–c) are periods whose rooms had different instructions, which a room field alone reproduces in synthetic data. (b) Per-unit exposure slopes (95% CI) and their random-effects summary: positive, but $p = 0.019$ against the required 0.01.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 11 ($O(n)$ vector spins) inside model 04’s superagent framing. Spin $\mathbf{s}_i$: agent i ’s day-mean statement embedding ($n = 32$). Fields: goal direction $\hat{g}$, agent field $\mathbf{h}_i$ (style, prior). Couplings J_{ij} act through room chat; E_{ij} : messages of j that i saw. p_c : share of a group’s statements in meaning cluster c . r_{ij} : pair alignment net of the fields; b : exposure slope (α, δ : pair, day effects). J_0 : mean-field coupling; L_n : Bessel ratio.

$$P(\{\mathbf{s}\}) \propto \exp\left(\sum_{i<j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j + \sum_i (h\hat{g} + \mathbf{h}_i) \cdot \mathbf{s}_i\right)$$

$$H_G = -\sum_c p_c \ln p_c, \quad \Delta H = H_{\text{rooms}} - \langle H_{\text{random}} \rangle$$

$$r_{ij}(t) = \alpha_{ij} + \delta_t + b \ln(1 + E_{ij}(t)) + \varepsilon_{ij}(t)$$

$$m = L_n(\beta(J_0 m + h)), \quad \text{critical point } \beta J_0 = n$$

Rooms make agents talk about the same things, which is obvious in hindsight, and the cleanest cases are rooms given different instructions: the operator’s field, not an emergent mind. The coupling residual is tiny and depends on the embedding; nobody would design channels around $+0.013$ cosine per e-fold. The one usable fact is where the shared dynamics live: day-to-day co-fluctuation sits inside rooms (correlation 0.06–0.66 within, about 0 across), so to read or damp herding, look at rooms, not labs. “Superagent” is, for now, a word for that. Whether room order is coupling or just a shared room drive is the NE15 holdout’s job, and it hasn’t run.

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RESULTS VS PREDICTIONS

Pending.

GOAL PERIODS IN DETAIL

NE15	...	pending (holdout)
G08	d	descriptive
G21	d	descriptive
G35	~	mixed
G36	×	failed
G37	×	failed
G38	~	mixed
G39	~	mixed
G40	✓	supported
G41	~	mixed
G42	~	mixed
G44	~	mixed
G51	~	mixed
NE32	–	n/a (not estimable as designed; descriptive numbers below)

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Pending.

ROUND 2

- **Essence.** effective superagents in Kolchinsky–Wolpert terms, running on a substrate of agents.
- **H01-R4** Individuality maxima pick out candidate superagents.
- **H01-R5** Superagents have positive group-level semantic information.
- **H01-R6** The agency is effective: it lives above the substrate.
- **H01-R7** Substrate independence.
- **H01-R8** The superagent does self-maintenance.

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